Local Civic Issue Tracker

Software Project Management Plan

Group 2 | Kazim Yigit Ablay and Michael Avila

November 4, 2026 | Reviewed by both members

This plan sets out the work remaining for Local Civic Issue Tracker through the final demonstration, together with the responsibilities, proposed schedule and responses to the risks that could affect delivery. Its tasks follow the submitted SRS and course requirements, with established responsibilities distinguished from new allocations and with estimates, availability and approvals recorded as pending inputs.

1 Overview

1.1 Project summary

1.1.1 Purpose scope and objectives

Local Civic Issue Tracker is a web application for following selected San Bernardino City Council issues across official records from 2023 onward, so users can trace a matter across meetings and consult the evidence behind its reported history. The initial delivery includes record preparation, connections, timelines, meeting overviews, summaries and search, with links to the original records and official videos. The submitted SRS v2.0 defines the scope and its conditional features.

1.1.2 Assumptions and constraints

Kazim and Michael form the development team. Both members expect to contribute approximately 10 hours per week each. The repository scaffold remains the documented implementation baseline, although the earlier prototypes provide feasibility evidence for selected examples. Development uses local instances and Kazim’s desktop GPU, with inference outputs shared with Michael and paid services outside the current plan.

1.1.3 Project deliverables

The planned deliverables are the application, its engineering documents, the verification and evaluation records, a shared demonstration dataset, and the final demonstration and retrospective. Annex A relates each output to the tasks that produce it and the evidence used to assess its completion. Conditional categorization and other stretch features remain outside the committed scope unless the team includes them, while the exact submission requirements remain pending confirmation.

1.1.4 Schedule and budget summary

The proposed schedule follows the tentative Fall 2026 calendar, which places cross-team review on November 17 and 19, a release candidate in the week of November 24, and final release and retrospective on December 1 and 3. Demonstrations fall within December 7–12. Section 5.2 gives the task sequence and dependencies, while effort estimates and the resulting capacity assessment remain pending.

Table 1. Milestone summary

Course checkpoint	Calendar window	Planned evidence
Cross-team engineering review	Nov 17/19	T12 available increment; T13 findings
Release candidate	Week of Nov 24	T14 integration; T15 assessment/corrections
Final release and retrospective	Dec 1/3	T17 release; T18 retrospective; T19 records
Demonstration and defense	Dec 7–12	T18 preparation, with the demonstration slot pending confirmation.

The calendar provides planning windows rather than confirmed upload deadlines, so Table 12 retains the SPMP submission date, an explicit CDR date and the final demonstration slot as pending confirmations.

1.2 Evolution of the plan

Michael is expected to continue maintaining the SPMP after its initial preparation, with the proposed review cycle following the weekly stand-up and any change to scope, estimates, availability or dependencies that affects delivery. Both members review changes to the delivery plan, with their purpose recorded in the revision history and the task, schedule and resource records reflecting the same revision so their dates and responsibilities remain consistent.

Revision history

Revision	Date and author	Summary
Initial plan	November 3, 2026 Kazim Yigit Ablay	Initial milestone, task, resource and risk plan based on the submitted SRS and course requirements. Team review and the inputs listed in Table 12 are pending.
Revision 1	November 4, 2026 Michael Avila	Recorded Michael's availability at approximately 10 hours per week and updated the capacity text. Confirmed Michael as lead for T07, T08, T10, T12, T15 and T16. Task-hour estimates and Kazim's confirmation of T03, T11, T14 and T17 are pending.

2 References

R1. CSE 6550. Software Project Management Plan assignment and grading rubric.

R2. IEEE Std 1058-1998. Course-provided SPMP section outline.

R3. Sommerville, I. Software Engineering, Chapter 23 Project Planning. Course lecture slides.

R4. Sommerville, I. Software Engineering, Chapter 22 Project Management. Course lecture slides.

R5. Local Civic Issue Tracker. Software Requirements Specification v2.0, November 1, 2026. Submitted copy.

R6. Local Civic Issue Tracker. ADR-0002 through ADR-0021 and repository development guidance. The ADRs retain their historical SRS v1.0 references.

R7. CSE 6550. Tentative Fall 2026 course schedule.

R8. Local Civic Issue Tracker. November 1 stand-up report and PDR peer-feedback record.

R9. CSE 6550. Week 10 Lecture Notes Key Takeaways, including AI-assisted implementation, planning and risk management.

3 Definitions

Effort is the time a person spends on a task, measured in person-hours, whereas duration is the calendar time over which that task occurs. A milestone provides a point for assessing progress, while a deliverable is the output supplied for review or use. A predecessor supplies an output needed by a later task. An open hour field means that the estimate is unrecorded, rather than that the work requires no effort.

4 Project organization

4.1 External interfaces

The instructor and course reviewers provide the requirements and feedback at the scheduled checkpoints, while official municipal sources supply the records on which the application depends. Intended users contribute to validation of its search and presentation. That validation remains planned in the SRS, so completion depends on participant evidence rather than the team’s assessment of its own implementation alone.

4.2 Internal structure

Both members contribute to development and review, with Michael leading detailed search design and Kazim leading T04–T06 and providing the outputs from local inference. Table 7 records those responsibilities alongside the proposed leads and reviewers for the remaining tasks, with approval of the new allocation still pending.

4.3 Roles and responsibilities

Table 2. Known responsibilities and remaining allocation

Member	Established responsibility	Pending review or input
Kazim	T04–T06, architecture and build direction, and local inference outputs.	Other proposed assignments and task-hour estimates.
Michael	Detailed search design and expected SPMP continuation.	Task-hour estimates.
Both	Joint review and course participation.	Delivery priorities, proposed allocation and capacity assessment.

5 Managerial process plans

5.1 Start up plan

5.1.1 Estimation plan

Effort estimates draw on the outputs of each task and the relevant prototype or development experience, including the learning, implementation, review, integration, corrections and documentation needed to produce a usable result. Code production accounts for only part of that work. Shared effort is divided by contributor and counted once in the total, while the numerical estimates remain pending team input.

The proposed dates describe working windows rather than effort, so their feasibility depends on the demand assigned to each person, the predecessor outputs and access to shared inference resources. The capacity assessment remains pending until the estimates are recorded,with any contingency allowance requiring an explicit decision rather than an assumed percentage.

5.1.2 Staffing plan

The project is staffed by Kazim Yigit Ablay and Michael Avila, with approximately 10 hours per week recorded for Kazim and Michael. Any reductions for particular weeks are also pending, and no additional staff are included in the current plan.

5.1.3 Resource acquisition plan

The project uses existing development equipment and Kazim’s RTX 4070 Ti for local inference. Outputs are shared with Michael, although the detailed transfer arrangement and practical runtime remain unresolved. A change to paid services would follow a separate decision about the technical limitation it addresses and its effect on cost and delivery.

5.1.4 Project staff training plan

Setup and learning effort are included in the tasks that need them, so the same activity is not counted again as a separate training allocation. T02 covers the build and runtime setup, T06 covers the local-model investigation, and each implementation task includes the learning needed for its component, with their hour estimates pending.

5.2 Work plan

5.2.1 Work activities

Tables 3–5 describe the planned outputs and completion evidence for the remaining work, drawing on the earlier feasibility experiments without adopting those prototypes as an implementation baseline. The allocation in Table 7 distinguishes confirmed, retained and proposed responsibilities, with the hour fields open because the capacity assessment depends on the estimates supplied by both contributors.

Table 3. Foundation and component work

ID and activity	Output and completion evidence
T01 Scope and remaining decisions	Both members establish the selected demonstration scope and record the decisions concerning identity, recovery, reprocessing and quality evaluation. The decision record identifies the agreed priorities and any unresolved items before dependent implementation begins. Basis: OQ-01, OQ-04–OQ-08.
T02 Build and runtime verification	This task establishes the runnable Java and Maven baseline and the minimum service setup. Recorded build and startup results identify what ran successfully and distinguish configuration checks from functional verification. Basis: LIM-04, LIM-05; V-27.
T03 Contracts and data design	The design defines the records, analysis and search boundaries, their APIs and the persistent representation needed for the first increment. The design review covers owner-controlled access, source references and representative requests and responses. Basis: LIM-06–LIM-09; DR-01–DR-13.
T04 Record acquisition and storage	The records service discovers and retrieves selected official inputs at configured intervals and request rates, retaining their original files, identities and meeting relationships, including earlier source versions. Completion evidence compares files and metadata with official examples and covers discovery, request-rate and storage checks. Basis: FR-01, FR-10, FR-11; DR-01–DR-06, DR-11, DR-16; V-16, V-24; FR-47, LIM-12.
T05 Native extraction and OCR	Readable content is prepared through native extraction where adequate, with OCR used for the scanned pages or graphics that need it. Comparisons with source pages establish extraction adequacy and preservation of the original-record association, including cases where an existing text layer is incomplete. Basis: LIM-10; FR-12, FR-18, FR-19, FR-42; V-01, V-09, V-16, V-18.
T06 Model feasibility and reference examples	The investigation assesses local whole-meeting inference and validation of usable model output, with reviewed development references retained separately from final evaluation cases. The record describes capacity limits, observed output and rubric decisions, while numerical acceptance targets follow development evidence before final evaluation. Basis: AS-03–AS-05; COM-10–COM-12; V-10, V-19, V-26.

Table 4. Analysis search and integration work

ID and activity	Output and completion evidence
T07 Evidence and issue connections	Analysis represents source evidence, developments and issue associations, creating an issue where a supported matter has no existing issue and leaving associations unset when support is insufficient. Reviewed examples cover new and existing matters, stated and inferred connections, and the passages supporting each association. Basis: FR-03, FR-21–FR-24, FR-41; DR-07, DR-08; V-03, V-19; FR-46.
T08 Summaries, meeting overviews and issue histories	Analysis produces summaries, agenda and outcome overviews, and dated issue histories with their qualifications, corrections and original-source or video links. A review against the official records covers unavailable outcomes, date meaning, uncertainty and treatment of later sources. Basis: FR-04–FR-09, FR-25–FR-32, FR-38; V-04–V-08, V-10, V-20, V-21.
T09 Search implementation	Search uses the agreed owner APIs and initial test data for issues, records, meetings and their selectable combinations. Detailed matching and ranking remain Michael-led. Reference queries and selection combinations provide the initial checks, followed by integration with published data under T14. Basis: FR-02, FR-36, FR-37; V-02, V-23.
T10 Frontend and API routing	The frontend provides views and API routing for search, inspection, timelines and evidence navigation, with the published collection’s coverage and the uncertainty of inferred connections visible to users. Checks cover selected result types, coverage information, uncertainty, source and video links, safe display and keyboard use. T14 completes live integration. Basis: FR-07, FR-08; NFR-02, NFR-03; V-07, V-08, V-11, V-12; FR-22, FR-23, FR-29–FR-32, FR-36, FR-37, FR-43–FR-45.
T11 Coordinator controls	The coordinator provides durable run admission, whole-meeting grouping, preparation and analysis progression, failure blocking and recovery contracts. Controlled overlapping launches and restart cases are exercised with service doubles before the corresponding live checks under T14 and T16. Basis: FR-13–FR-17, FR-20, FR-34, FR-35; DR-13; NFR-12; V-17, V-18.
T12 Cross-team review package	The review package contains the available increment, setup instructions, test inputs and an explicit account of unfinished analysis or views using test data. A reproduced setup identifies the functions another team can exercise and the limits of the current increment. Basis: Course cross-team review; COM-05; V-27.

Table 5. Review release and recurring work

ID and activity	Output and completion evidence
T13 Cross-team engineering review	Both members participate in the course engineering review and retain findings about reproducibility, integration and correctness. The review record identifies observations for correction without treating participation as a passed acceptance test. Basis: Tentative course calendar and cross-team review assignment.
T14 Full run and publication integration	This task integrates the live records, OCR, analysis, search, frontend and coordinator flows, including publication of a completed run and retention of the prior view after failure. End-to-end checks cover completed, unfinished and failed runs, search of published data and retained source provenance. Basis: FR-33–FR-35; NFR-04, NFR-10–NFR-13; V-13, V-17, V-22, V-23.
T15 Review corrections and candidate assessment	The candidate assessment addresses prioritized findings from cross-team review and integration, with remaining defects recorded alongside their effect on delivery. Affected checks are repeated and a candidate decision records the available evidence and unfinished work. Basis: Course release-candidate window; V-14.
T16 Integrated verification and evaluation	Integrated evaluation covers the agreed reference cases, recovery, security, browser behavior and user-facing validation, together with regression corrections. Observed results are recorded against the SRS checks and previously agreed targets. Conditional categories are evaluated only when included by team agreement. Basis: V-01–V-27, V-29, V-30; PR-02–PR-05; NFR-01–NFR-18.
T17 Final release and restoration	The release package combines the application, coordinated database and files, current engineering and test records, known limitations and demonstration inputs. A restored dataset and repeated demonstration steps provide evidence of reproducibility and package completeness. Basis: DR-17; V-28; course final-release window.
T18 Retrospective and defense preparation	Both members prepare the team demonstration and retrospective, while each prepares their own understanding and any required individual reflection or defense. Rehearsal uses the release and its recorded limitations. The course prohibition on AI during the individual oral defense applies. Basis: Course December 1/3 and December 7–12 windows.
T19 Engineering documentation and coordination	The recurring documentation work maintains the SRS, SPMP, ADRs and necessary design and verification records as development changes. Milestone reviews and weekly progress records identify the current state. The task estimate includes this recurring coordination and documentation effort. Basis: COM-01–COM-13; V-14, V-15; weekly stand-up.

5.2.2 Schedule allocation

Table 6 gives the proposed task sequence, with dependencies attached to the outputs a later task needs rather than to the passage of a date alone. Search, frontend and coordinator development can begin against agreed contracts and test data before T14 completes their live integration, although the cross-team review uses only the available increment with its documented limitations rather than implying final acceptance.

Table 6. Proposed task windows and predecessor outputs

Task	Start	Finish	Predecessors
T01 Scope and remaining decisions	Nov 04	Nov 05	None
T02 Build and runtime verification	Nov 04	Nov 05	None
T03 Contracts and data design	Nov 05	Nov 06	T01, T02
T04 Record acquisition and storage	Nov 07	Nov 10	T03
T05 Native extraction and OCR	Nov 11	Nov 12	T04
T06 Model feasibility and reference examples	Nov 13	Nov 15	T05
T07 Evidence and issue connections	Nov 16	Nov 18	T06
T08 Summaries, meeting overviews and issue histories	Nov 19	Nov 21	T07
T09 Search implementation	Nov 07	Nov 14	T03
T10 Frontend and API routing	Nov 07	Nov 14	T03
T11 Coordinator controls	Nov 07	Nov 14	T03
T12 Cross-team review package	Nov 15	Nov 16	T05, T09, T10, T11
T13 Cross-team engineering review	Nov 17	Nov 19	T12
T14 Full run and publication integration	Nov 22	Nov 23	T08, T09, T10, T11
T15 Review corrections and candidate assessment	Nov 24	Nov 25	T13, T14
T16 Integrated verification and evaluation	Nov 26	Nov 29	T06, T15
T17 Final release and restoration	Nov 30	Dec 01	T16
T18 Retrospective and defense preparation	Dec 02	Dec 06	T17
T19 Engineering documentation and coordination	Nov 03	Dec 06	None

T19 spans the development period and supplies updated records at each checkpoint, while component tasks include their own review and testing before T16 brings the final integrated evaluation together. The proposed T15 window continues through November 25, allowing candidate corrections after the start of the course’s release-candidate week and leaving readiness at that checkpoint dependent on the results actually available.

Figure 1. Proposed task schedule by week

Task	Nov 3–9	Nov 10–16	Nov 17–23	Nov 24–30	Dec 1–7
T01 Scope and remaining decisions	Work				
T02 Build and runtime verification	Work				
T03 Contracts and data design	Work				
T04 Record acquisition and storage	Work	Work			
T05 Native extraction and OCR		Work			
T06 Model feasibility and reference examples		Work			
T07 Evidence and issue connections		Work	Work		
T08 Summaries, meeting overviews and issue histories			Work		
T09 Search implementation	Work	Work			
T10 Frontend and API routing	Work	Work			
T11 Coordinator controls	Work	Work			
T12 Cross-team review package		Work			
T13 Cross-team engineering review			Work		
T14 Full run and publication integration			Work		
T15 Review corrections and candidate assessment				Work	
T16 Integrated verification and evaluation				Work	
T17 Final release and restoration				Work	Work
T18 Retrospective and defense preparation					Work
T19 Engineering documentation and coordination	Work	Work	Work	Work	Work

The shaded cells identify the weeks in which a task is planned, while Table 6 provides its exact dates and predecessor IDs. The figure represents calendar windows rather than effort or percentage completion.

The December 1 release target follows T17 and depends on the preceding integration, correction and evaluation results, with the broader course windows recorded in section 1.1.4. Outstanding estimates and scope confirmations affect whether that sequence is achievable, so readiness is assessed from the required outputs and the evidence that supports their completion.

5.2.3 Resource allocation

All tasks are assigned. Each estimate includes the hours contributed by both members to implementation, review and support, with the task total representing their sum and the numerical fields pending.

Table 7. Task allocation and effort fields

Task	Lead and status	Review / support	Kazim h	Michael h	Total h
T01	Both (confirmed)	Joint decision	Open	Open	Open
T02	Kazim (retained)	Michael	Open	Open	Open
T03	Kazim (confirmed)	Michael	Open	Open	Open
T04	Kazim (confirmed)	Michael	Open	Open	Open
T05	Kazim (confirmed)	Michael	Open	Open	Open
T06	Kazim (confirmed)	Michael	Open	Open	Open
T07	Michael (confirmed)	Kazim	Open	Open	Open
T08	Michael (confirmed)	Kazim	Open	Open	Open
T09	Michael (retained)	Kazim	Open	Open	Open
T10	Michael (confirmed)	Kazim	Open	Open	Open
T11	Kazim (confirmed)	Michael	Open	Open	Open
T12	Michael (confirmed)	Kazim	Open	Open	Open
T13	Both (retained)	Joint participation	Open	Open	Open
T14	Kazim (confirmed)	Michael	Open	Open	Open
T15	Michael (confirmed)	Kazim	Open	Open	Open
T16	Michael (confirmed)	Kazim	Open	Open	Open
T17	Kazim (confirmed)	Michael	Open	Open	Open
T18	Both (retained)	Individual defense preparation	Open	Open	Open
T19	Both (retained)	Michael: SPMP; Kazim: design	Open	Open	Open

A task lead coordinates its work and completion evidence, while the other member may contribute to implementation or review rather than leaving the entire task to one person. Michael is confirmed to coordinate T16, with each member reviewing the other’s implementation and with T19 retaining Michael’s SPMP role and proposing design updates for Kazim. Each member remains responsible for their individual defense.

Table 8. Available hours and workload check

Planning week	Kazim capacity	Michael capacity	Assigned demand
Nov 3–9	Approximately 10 h	Approximately 10 h	Open
Nov 10–16	Approximately 10 h	Approximately 10 h	Open
Nov 17–23	Approximately 10 h	Approximately 10 h	Open
Nov 24–30	Approximately 10 h	Approximately 10 h	Open
Dec 1–7	Approximately 10 h	Approximately 10 h	Open

At a steady 10 hours per week, Kazim and Michael have approximately 40 hours each of planning capacity over the four weeks from November 3 to December 1. The December 1–7 row belongs to a later full-week bucket and is outside that subtotal. Holiday reductions and defense-week availability remain pending, while team capacity and assigned demand cannot yet be compared because the estimates are open.

5.2.4 Budget allocation

No paid services or new equipment purchases are included in the current plan, which uses existing hardware and treats person-hours as the principal resource allocation to assess. A monetary spending limit has not been established.

5.3 Control plan

The proposed control approach uses the task register, SRS records and weekly reporting already associated with the project, so the management record remains connected to the work it describes. Approval of these practices, the resource allocation and the capacity assessment remains pending team review.

5.3.1 Requirements control plan

A proposed requirement change is recorded with the affected SRS IDs, its reason and its implications for design, verification, effort and delivery before it enters the planned scope. Both members review the change. Their decision appears in the revision history and traceability records, while any unresolved proposal retains that status until a decision is recorded.

5.3.2 Schedule control plan

The proposed weekly review compares completed outputs and remaining work with the schedule, including tasks that are waiting for a predecessor or the GPU. Once estimates are available, that comparison also includes the demand assigned to each person and their remaining capacity. Dates and dependencies are revised together, while any reduction in required scope or change to course expectations requires explicit agreement.

5.3.3 Budget control plan

The current budget contains no planned paid-service expenditure, although a proposed purchase or hosted-model change would be assessed against its technical purpose, expected cost and effect on delivery before either member commits to it. Any approved expenditure becomes part of the budget record alongside the existing hardware and effort allocation.

5.3.4 Quality control plan

Task completion criteria and the applicable SRS scenarios provide the basis for integration decisions, with failed and unfinished checks recorded alongside their corrective work. For model-assisted output, the review also examines the official evidence because a valid response structure does not establish that its claims are supported. Sections 6.4 and 7.2 describe acceptance and verification in more detail.

5.3.5 Reporting plan

The weekly stand-up reports completed work, immediate plans, individual contributions, deliverable status, risks and AI assistance, while the proposed plan review accompanies that report. Both members confirm the work attributed to them. Partial increments are described by the functions they support, including any views that still use test data, and changed commitments appear in the task register.

5.3.6 Metrics collection plan

The project records the status and effort of its tasks, the variance from milestones, unresolved defects and risks, and the verification status of the SRS requirements. Assigned weekly demand is the sum of the hours contributed by both members, with spare capacity representing the difference between availability and that demand. The calculations remain pending. Model results retain the SRS measures and denominators, and a passed verification status requires a recorded check that supports that conclusion.

5.4 Risk management plan

Tables 9 and 10 contain the 19 identified risks to the project and product, with ratings that express qualitative judgments about their likelihood and consequences rather than measured probabilities. The responses and owners remain pending review. The proposed weekly review gives early attention to High/Serious entries and to dependencies that could block the next task, with reassessment when a warning indicator appears.

Likelihood ranges from Very low to Very high through Low, Moderate and High, while consequence is rated Insignificant, Tolerable, Serious or Catastrophic, following Chapter 22. Avoidance reduces the likelihood of a risk, minimization reduces its impact, and contingency describes the response if it occurs. Type identifies the cause. The affected area identifies whether its consequences concern the project or product, while a current defect is recorded as an issue even if its future effects remain relevant to the register.

Table 9. Proposed risk assessment

ID	Risk and rating basis	Type and effect	Likelihood	Consequence
RK-01	Remaining work exceeds available hours before the course reviews or final release. Kazim and Michael have approximately 10 hours per week, while task estimates remain pending.	Estimation and people Project	High	Serious
RK-02	Portal changes or missing source inputs prevent reliable collection. Collection depends on external official records and the imported selection does not represent a complete archive.	Technology and requirements Project and product	Moderate	Serious
RK-03	Extraction omits or misreads content needed for analysis. PR-01 established feasibility for selected examples, while broader completeness and difficult image content remain unverified.	Technology Product	High	Serious
RK-04	Whole-meeting analysis exceeds local memory, context or practical runtime limits. The GPU is available, but representative whole-meeting capacity and the production model configuration remain unverified.	Technology and estimation Project and product	Moderate	Serious
RK-05	Analysis produces unsupported claims or incorrect issue connections. Supported summaries and inferred links are central functions, while their final evaluation evidence is still planned.	Technology and requirements Product	High	Serious
RK-06	Restart or publication errors overlap runs or expose incomplete results. The selected exclusion, recovery and publication behavior remains to be demonstrated in application execution.	Technology Project and product	Moderate	Serious
RK-07	Service and data integration requires more rework than the schedule allows. The documented baseline is a scaffold, and integration spans Java services, Python OCR, inference, storage and the browser frontend.	Technology and estimation Project	High	Serious
RK-08	Reviewed examples or intended-user feedback arrive too late for evaluation. The SRS identifies reference preparation and intended-user validation as planned work without completed participant outcomes.	People and requirements Project and product	Moderate	Serious

ID	Risk and rating basis	Type and effect	Likelihood	Consequence
RK-09	Selected dependencies, runtimes or API calls are unavailable or incompatible, including plausible AI suggestions. The runnable baseline remains unverified, and plausible dependency suggestions may still refer to unavailable artifacts or unsupported methods.	Tools and technology Project	Moderate	Tolerable
RK-10	Generated code introduces security defects or exposes sensitive configuration. Imported content and model responses cross application trust boundaries, creating opportunities for unsafe handling in either manual or generated code.	Tools and technology Project and product	Moderate	Serious
RK-11	AI-produced code is accepted without sufficient human review or verification. Time pressure can shift effort away from understanding and reviewing AI-assisted changes before integration.	People and tools Project and product	Moderate	Serious
RK-12	Unavailable local inference hardware or delayed output sharing blocks dependent work. The planned inference resource is Kazim’s desktop GPU, with outputs shared to Michael rather than an established remote endpoint.	People and technology Project	Moderate	Serious
RK-13	Database and retained files diverge or cannot be restored together after sharing or local data loss. Files and SQL references are stored separately, while DR-17 and ADR-0012 require matching snapshots with restoration checks.	Technology Project and product	Moderate	Serious
RK-14	New, corrected or repeated source inputs leave stale histories or incorrect source-version associations. The SRS requires updates to current history while retaining source versions, but detailed identity and reanalysis rules remain open.	Technology and requirements Product	Moderate	Serious
RK-15	Search and evidence presentation fail intended-user tasks despite usable stored records. Users need to find information without known dates and interpret coverage, dates and inferred links. The corresponding validation remains planned.	Requirements and technology Product	Moderate	Serious
RK-16	Evaluation overstates quality because final cases overlap development work or lack representative, well-supported reference judgments. Section 11.1 separates development and final cases and requires reviewed judgments, recorded configuration and explicit denominators.	People and requirements Project and product	Moderate	Serious
RK-17	Late scope, acceptance or course-date clarification forces rework or leaves required delivery work unplanned. The SRS contains conditional and unresolved items, while course dates remain tentative and some submission deadlines are unconfirmed.	Requirements and organizational Project and product	Moderate	Serious

ID	Risk and rating basis	Type and effect	Likelihood	Consequence
RK-18	Overlooked third-party use conditions require dependency replacement or release-package changes. The stack includes third-party libraries and model components whose final production selections remain open. No licence conflict has been identified.	Organizational and requirements Project	Low	Tolerable
RK-19	Processing consumes shared resources or holds locks that stall interactive browsing and search. Services share local resources and one PostgreSQL instance, while NFR-10 and NFR-13 require browsing during processing.	Technology Product	Moderate	Serious

The register links each response to existing tasks and a proposed owner, with the effort included in those estimates once the allocation is reviewed. Its entries describe possible failures arising from the selected design and current uncertainty, without establishing that those failures have occurred. RK-18 concerns a review of third-party use conditions, with no licence conflict currently identified.

Table 10. Proposed responses ownership and warning indicators

ID	Strategy and response	Warning indicator and proposed owner
RK-01	The team compares task estimates with weekly capacity while retaining time for review, integration and testing. If the work exceeds that capacity, a revised scope or schedule is considered with the appropriate reviewers before the delivery commitment changes. Strategy: Minimization, contingency. Tasks: T01, T15, T19.	An overloaded week, missed prerequisite or growing unfinished work at a checkpoint. Proposed owner: Both members.
RK-02	Collection checks cover representative portal relationships, expected inputs and transcript availability, with requests and retries governed by LIM-12. Discovery failures remain distinct from a source with no new records. A changed access path leads to investigation and review of any proposed scope change. Strategy: Minimization, contingency. Tasks: T04, T05, T19.	Missing expected inputs, retrieval failures, changed portal responses or request limiting. Proposed owner: Kazim.
RK-03	Representative native and OCR outputs are compared with source pages, including partially populated text layers. Failed examples lead to preparation changes and repeated checks, while a change to the supported input scope follows the requirement-change process. Strategy: Minimization, contingency. Tasks: T05, T06, T16.	Relevant source text absent, incorrect labels or failed adequacy checks. Proposed owner: Kazim.
RK-04	T06 tests representative long meetings and records the model configuration and practical limits. If the whole-meeting approach proves unsuitable, alternatives and their design implications are reviewed before a change is adopted. A paid service remains a separate decision. Strategy: Avoidance, contingency. Tasks: T06, T19.	Memory errors, unusable/truncated results or runtime that prevents scheduled work. Proposed owner: Kazim.
RK-05	Reference review covers source support, similarly named but different matters, expected new issues and missed associations. Output validation preserves uncertainty, while acceptance targets follow development evidence. Unsupported results lead to analysis corrections before acceptance is claimed. Strategy: Minimization, contingency. Tasks: T06, T07, T08, T16.	Unsupported claims, incorrect associations or lost qualifications during source review. Proposed owner: Michael.

ID	Strategy and response	Warning indicator and proposed owner
RK-06	Controlled checks exercise overlapping launches, interrupted service calls and failed runs. Later work remains pending while a failure is unresolved, and the prior published view stays available. Ambiguous remote execution is resolved before a retry so recovery does not silently skip meetings or duplicate work. Strategy: Minimization, contingency. Tasks: T11, T14, T16.	A second run starts, saved state disagrees or partial results become visible. Proposed owner: Kazim.
RK-07	Early contract review and incremental integration expose boundary problems before the final run. Live owner APIs replace test data during integration, with saved outputs checked across services. A changed contract leads to coordinated replanning and regression checks. Strategy: Minimization, contingency. Tasks: T02, T03, T12, T14, T15.	Fixture checks pass while live calls, persistence or setup fail. Proposed owner: Kazim.
RK-08	Reference review and intended-user feedback are arranged early enough for corrections, with development cases kept separate from final evaluation. If the required evidence is unavailable, the evaluation record states that limit and any revised approach follows team review. Strategy: Avoidance, minimization. Tasks: T01, T06, T16, T19.	No reviewers or usable reference judgments when dependent evaluation starts. Proposed owner: Both members.
RK-09	Suggested dependencies, supported runtimes and API calls are checked against official documentation and actual build or startup results, including Python, OCR and model setup. Unsupported choices are replaced and affected checks repeated. Changes to the selected stack or use of paid alternatives remain separate decisions. Strategy: Avoidance, contingency. Tasks: T02, T03, T05, T06, T19.	An artifact cannot resolve, a runtime cannot start or a required method is absent or incompatible. Proposed owner: Task implementer.
RK-10	The review covers the source-instruction boundary, browser display, output validation, credentials and owner-controlled data access in both manual and generated code. Controlled malicious inputs exercise those safeguards. Identified defects are corrected before release, and an exposed credential is rotated if an exposure occurs. Strategy: Avoidance, minimization, contingency. Tasks: T03, T10, T16, T19.	Injected actions, script execution, unsafe data handling or a committed secret. Proposed owner: Implementer and reviewer.
RK-11	Implementation increments remain small enough for human review against the relevant SRS and ADRs. The review covers the diff and meaningful test results, with corrections and rejected AI suggestions recorded. Its effort is included in the task estimate. Strategy: Avoidance, minimization. Tasks: T01, T14, T16, T19.	A contributor cannot explain a change or its claimed checks have no results. Proposed owner: Task implementer and reviewer.
RK-12	Shared inference outputs retain their input and configuration identity, with verified examples and setup information available to both developers. If the machine or contributor becomes unavailable, independent work continues with suitable saved examples while dependent tasks are replanned. Replacement hardware or paid inference is not assumed. Strategy: Minimization, contingency. Tasks: T06, T07, T08, T14, T19.	A dependent task is waiting for a required inference run, unavailable machine or missing shared output. Proposed owner: Kazim; Michael identifies needed outputs.
RK-13	Dataset copies combine the database state with its matching retained files, and restoration is checked under the recipient’s configured storage location. Recoverable copies preserve a verified reference set. An incomplete copy is replaced with a matching set rather than merged with divergent databases or treated as authority to resume active jobs. Strategy: Avoidance, minimization, contingency. Tasks: T03, T04, T12, T17.	A restored record has missing files/results, mismatched versions or unusable storage references. Proposed owner: Kazim; Michael checks recipient restore.

ID	Strategy and response	Warning indicator and proposed owner
RK-14	T01 and T03 cover the relevant identity and reanalysis decisions. Checks then exercise repeated inputs, explicit corrections and later sources describing earlier developments, including their effects on published results and source-version support. Incorrect updates are corrected and evaluated before publication, while earlier source evidence remains retained. Strategy: Avoidance, minimization, contingency. Tasks: T01, T03, T04, T07, T08, T14, T16.	A correction changes source evidence but not the displayed account, or repeated imports create duplicate/inconsistent identities. Proposed owner: Michael; Kazim supplies versioned inputs.
RK-15	Agreed user tasks exercise ordinary wording, selected result types, coverage, source access and uncertainty. Missing imports are distinguished from search misses. Where users cannot complete or interpret a task, the matching or presentation is revised within Michael-led design and the affected checks are repeated. Strategy: Avoidance, minimization, contingency. Tasks: T09, T10, T13, T16.	Relevant published content is not found, or users mistake collection coverage, inferred links or displayed date meaning. Proposed owner: Michael; Kazim reviews source/evidence cases.
RK-16	Reviewed judgments and ambiguous cases define the evaluation basis, with final cases kept separate from prompt tuning and relevant difficult inputs represented within the selected scope. Configuration, exclusions and denominators accompany the results. Overlap or weak judgments lead to corrected references and additional final cases where needed, with limitations reported explicitly. Strategy: Avoidance, minimization, contingency. Tasks: T06, T16, T19.	Final cases influenced tuning, judgments lack source support, or reported metrics omit misses, exclusions or undefined denominators. Proposed owner: Michael; both review reference judgments.
RK-17	The plan records the selected delivery scope and course dates, with conditional features separated from required behavior. A change is assessed against tasks, verification and effort before priorities or delivery commitments are revised. Required behavior is not removed without agreement, and an extension is not assumed. Strategy: Avoidance, minimization, contingency. Tasks: T01, T15, T19.	An unresolved scope decision blocks implementation, an unplanned feature is added or confirmed dates conflict with the working plan. Proposed owner: Both members.
RK-18	Selected code, model weights and bundled data receive a review of their applicable use and redistribution conditions before packaging, with required notices retained. If a condition conflicts with intended use, an alternative component or package arrangement is reviewed. Public availability alone is not treated as permission for every use. Strategy: Avoidance, contingency. Tasks: T02, T05, T06, T17, T19.	Use conditions or required notices are missing from the review, or a selected component cannot be included as planned. Proposed owner: Both members; each checks selected components.
RK-19	Browsing and search are exercised against published results during a representative processing run. Long waits and resource or lock contention lead to review of transaction scope and processing resource use, followed by repeated checks. Interactive requests remain independent of run completion under the existing SRS requirements. Strategy: Avoidance, minimization, contingency. Tasks: T03, T09, T11, T14, T16.	Published-data requests wait for a processing step or become unusable while the batch run is active. Proposed owner: Kazim; Michael checks interactive requests.

Risk-response effort is included in the referenced tasks, or in a separately identified task if the team later divides that work, with shared effort counted once. The proposed weekly review records whether each action is planned, underway or verified and whether the assessment has changed. Contingency allowances and scope changes remain subject to an explicit decision.

5.5 Closeout plan

T17 brings together the release, restoration instructions, engineering records, evaluation evidence and known limitations, while T18 covers the demonstration and retrospective and each member prepares their own defense. The final handoff describes the functions delivered, unresolved defects and deferred work, with the exact course artifacts and upload dates retained as pending confirmations in Table 12.

6 Technical process plans

6.1 Process model

The proposed process combines course milestones with incremental development, beginning each increment with the relevant SRS behavior, ADR constraints and observable completion criteria before implementation, review and integration. The path then expands in another increment. Planning, implementation and verification can occur in stages within one AI tool or through separate roles where the task benefits from that separation, with a team member retaining responsibility for scope and review.

6.2 Methods tools and techniques

The backend uses Java and Spring Boot for records, search, analysis and coordination, while Python provides OCR and React, TypeScript and Vite form the frontend. The Java builds are managed through Maven. PostgreSQL holds the queryable data and file references in schemas owned by the services, with JPA providing domain persistence for records and analysis and with inference returning output to its caller for validation and storage. Detailed search design remains Michael-led, while the production inference components and the conditional Spring Batch choice retain their planned feasibility, recovery and non-overlap checks.

6.3 Infrastructure plan

Development uses separate local instances, with Kazim’s desktop providing inference outputs for Michael and those shared results retaining the input and configuration identity of the run that produced them. The sharing design combines a database snapshot with its matching files and a checked restore under the recipient’s configured storage location. A copied dataset does not imply resumption of active execution state. Setup and access details reside in development instructions and configuration, with credentials excluded from version control.

6.4 Product acceptance plan

Acceptance follows the current SRS, its user stories and the applicable V-01 through V-30 scenarios, with T06 establishing the reviewed development references and the rubric for a supported association. Numerical targets are agreed from those results before final evaluation uses separately retained cases under a recorded configuration, with the reported precision, recall, evidence presence and evidence support accompanied by their denominators and any ambiguous or excluded cases.

During T16 and T17, both members review required behavior, failures, source support, restoration and demonstration readiness, with each applicable check assigned a recorded pass, fail or unfinished status and any defect given a disposition. Conditional categorization is evaluated only if included, and a demonstration or successful model call contributes evidence without establishing acceptance of the whole system.

7 Supporting process plans

7.1 Configuration management plan

The Git repository holds selected source code, configuration, development instructions and maintained engineering documents, while private working artifacts remain outside that set. Under the proposed review practice, each change is associated with its task and relevant SRS or ADR before its diff and checks are reviewed for integration. Evaluation results retain the source, dataset, model, prompt and configuration identities needed to understand the run. A specific branch policy or CI service is not selected.

7.2 Verification and validation plan

Component tasks check their own behavior and output, while T03 and T14 examine the contracts and APIs through which the services obtain data from its owners. T11, T14 and T16 cover chronology, recovery, run exclusion and publication, with extraction and model output compared with official evidence under T05, T06 and T16. Browser, keyboard and security checks appear in T10 and T16, and T17 includes restoration. Intended-user validation remains planned until participant evidence is recorded.

A verification record identifies the SRS scenario, inputs, configuration, expected behavior, observed outcome, date, reviewer and any defect so an individual result can be assessed against its evidence. The review distinguishes a broad task allocation from a single test and examines the assertions and failure cases in AI-assisted tests, including interactions where an isolated component test cannot demonstrate the required behavior.

7.3 Documentation plan

The engineering set contains the SRS, SPMP, ADRs and the design, setup, verification and delivery records needed to describe the implementation and support its review. Michael is expected to maintain the SPMP, with the proposed allocation placing design updates with Kazim and component records with their contributors. Revisions identify material changes, and selected exports remain associated with their available editable sources. Annex A identifies the main outputs.

7.4 Quality assurance plan

Quality assurance examines whether the agreed process is followed and whether reported completion has supporting evidence, complementing the product checks through review of traceability, exceptions and corrective work across the project. Sections 5.3.4 and 7.2 describe those product checks. The two-person review arrangement is part of the team’s work rather than a separate assurance department, and documentation or isolated tests are assessed according to the evidence they provide.

7.5 Reviews and audits

The proposed routine review accompanies the weekly stand-up and precedes integration of a substantial increment, with further checkpoints at cross-team review, candidate assessment and final release. Each review considers the scope, interfaces, requirement coverage, risks and reproducibility of the work, with its feedback and resulting actions retained in the project record. The November calendar event is cross-team review, with an explicit CDR date pending confirmation.

7.6 Problem resolution plan

A problem record identifies the affected task or requirement, the evidence needed to reproduce it, its impact and the person responsible, with priority reflecting its effect on correctness and delivery. Closure follows correction and repetition of the relevant checks. Any resulting change to future work appears in the schedule or risk record, with the proposed tracking format remaining a small shared register and with production recovery authority and reprocessing details covered by T01.

7.7 Subcontractor management plan

Subcontractor management is not applicable because no subcontracted delivery work is planned for the two-member team, although external libraries and any later paid service retain their technical, licensing and cost review.

7.8 Process improvement plan

The proposed progress review uses estimation differences, repeated defects and integration delays to adjust the next work package, with process changes recorded alongside their effect on the plan. This allows useful improvements to be assessed against their overhead. The final retrospective draws on actual experience to explain what worked, where rework arose and what could improve in a later iteration.

8 Additional plans

Codex assisted with organizing the course and project material, comparing the plan with the SRS and ADRs, and reviewing its structure, wording and layout, while proposed allocations and risk assessments remain for human review. Kazim provided his availability and resource direction and assigned T04–T06 to himself. This revision checks that the rewritten plan retains its responsibilities, dates, references and ratings, with final team review and its corrections still pending.

The course calendar prohibits AI assistance during the individual oral defense, and each member is responsible for explaining their contribution and the design and evidence supporting the project.

Annex A Deliverables and pending review

Table 11. Deliverable register

Output	Related tasks and responsibility	Completion evidence and checkpoint
Available application increment	T02–T14. Allocation status is recorded in Table 7.	Reproducible setup instructions and tested functions, with unfinished features identified in the cross-team review record.
Release candidate and final application	T14–T17. Kazim is confirmed for integration and release, and Michael for candidate and evaluation coordination.	Candidate assessment, verification outcomes and known limitations within the tentative November 24 and December 1 windows.
Engineering documents	T01, T03 and T19. Shared records, with Michael continuing the SPMP and Kazim confirmed for design updates.	Current SRS, SPMP, ADRs and required design and setup records with revision history and valid references.
Verification and evaluation record	T05–T17. Michael is confirmed as the T16 coordinator, with both members contributing and reviewing.	Recorded inputs, configuration, source judgments, results and defect dispositions linked to the applicable SRS scenarios.
Shared data and demonstration package	T17 and T18. Kazim is confirmed as package lead, with Michael checking restoration on his machine.	A checked restoration of the matching files and database, with reproducible demonstration steps.
Retrospective and individual defense	T18 and T19. Both members contribute, with each responsible for their individual work.	A prepared team demonstration, retrospective and individual understanding. Exact artifacts and deadlines remain pending confirmation.

Table 12. Pending planning inputs and approvals

Input	Current status
Task hours and availability	Task and contributor estimates are pending. Workload feasibility is unconfirmed.
Assignments and estimates	T04–T06 are assigned to Kazim. T04–T06 are assigned to Kazim and T07, T08, T10, T12, T15 and T16 to Michael. Leads for T03, T11, T14 and T17 await Kazim's confirmation.
Scope and course dates	Exact SPMP, CDR and upload dates, the final demonstration slot and delivery priorities remain pending confirmation. Conditional scope retains its SRS status.
Risk and process approval	Team review of the proposed ratings, owners, responses and management practices is pending.
Acceptance preparation	The detailed evaluation rubric, development-based numerical targets and remaining SRS decisions are pending. Targets precede the final evaluation.
Human review of AI assistance	Final team review, resulting corrections and any rejected assistance are pending.

Index

A separate index is not applicable. Numbered headings and captions identify the sections, tables and schedule figure.